

A SHORT PROOF THAT LINK-IRREGULAR TOURNAMENTS EXIST FOR EVERY ORDER AT LEAST SIX

DEEP BHATTACHARJEE

ABSTRACT. A digraph is link-irregular if no two of its vertices have isomorphic links. Bastien and Khormali conjectured that a link-irregular tournament on n vertices exists if and only if $n \geq 6$, and Chau proved it by substitution into a transitive tournament. We give a short different proof, by a construction that adds two vertices at a time.

1. INTRODUCTION

The link $L_D(v)$ of a vertex v in a digraph D is the subdigraph induced by the in- and out-neighbours of v , and D is *link-irregular* if $L_D(u) \not\cong L_D(v)$ for all distinct vertices u, v . Links of vertices in graphs were introduced by Ali, Chartrand and Zhang [1], and Bastien and Khormali [2] carried the notion over to digraphs. In a tournament every other vertex is a neighbour of v , so the link of v is the subtournament $T - v$. A tournament is therefore link-irregular exactly when its n vertex-deleted subtournaments are pairwise non-isomorphic.

Bastien and Khormali showed that there is no link-irregular tournament on $n \leq 5$ vertices (for $n = 5$ there are only four tournaments on four vertices, up to isomorphism, to serve as links), gave link-irregular tournaments on 6, 7 and 8 vertices, and found examples by computer search for every $n \leq 100$. Their construction adds one dominating vertex at a time and stops working at 9 vertices. They made the following conjecture ([2, Conjecture 6]; Conjecture 3.7 in the journal version [3]).

Conjecture 1. A link-irregular tournament on n vertices exists if and only if $n \geq 6$.

Here $n \geq 2$; the tournament with one vertex is link-irregular for want of a second vertex.

Chau [4] proved Conjecture 1. Call a tournament 2-strong if it is strongly connected and stays so after any one vertex is deleted. Chau showed that substituting link-irregular 2-strong tournaments into a transitive tournament gives a link-irregular tournament, and he found such tournaments of every order from 8 to 15 by computer search; together with the examples of Bastien and Khormali this covers every $n \geq 6$. Harte formalised Chau's proof in Lean [6] and showed that 7 is the least order of a link-irregular 2-strong tournament [5].

We give a second proof. Its only finite input is a pair of tournaments on six and seven vertices whose links can be told apart by hand.

Theorem 1.1. *For every $n \geq 6$ there is a link-irregular tournament on n vertices that has neither a source nor a sink.*

The proof is an induction from n to $n + 2$ (Lemma 2.1), started at $n = 6$ and $n = 7$ (Section 3). The new pair of vertices is a source and a sink of the old tournament joined by one reversed arc; this pair can be recognised inside every link, which is what makes the step work.

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Corresponding author: d.bhattacharjee@erl-forschung.de; ORCID [0000-0003-0466-750X](https://orcid.org/0000-0003-0466-750X).

We write $u \rightarrow v$ when the arc between u and v points from u to v , and $\text{sc}_T(v)$ for the score (out-degree) of v in a tournament T . A source of T is a vertex of score $|T| - 1$ and a sink is a vertex of score 0. Isomorphisms of tournaments preserve scores.

2. THE STEP FROM n TO $n + 2$

Let T be a tournament on a vertex set V with $|V| = n$. Let T^+ be the tournament on $V \cup \{s, t\}$ that contains T and has the arcs

$$s \rightarrow v \text{ and } v \rightarrow t \quad (v \in V), \quad t \rightarrow s$$

(Figure 1). In T^+ we have $\text{sc}(s) = n$, $\text{sc}(t) = 1$ and $\text{sc}(v) = \text{sc}_T(v) + 1$ for $v \in V$.

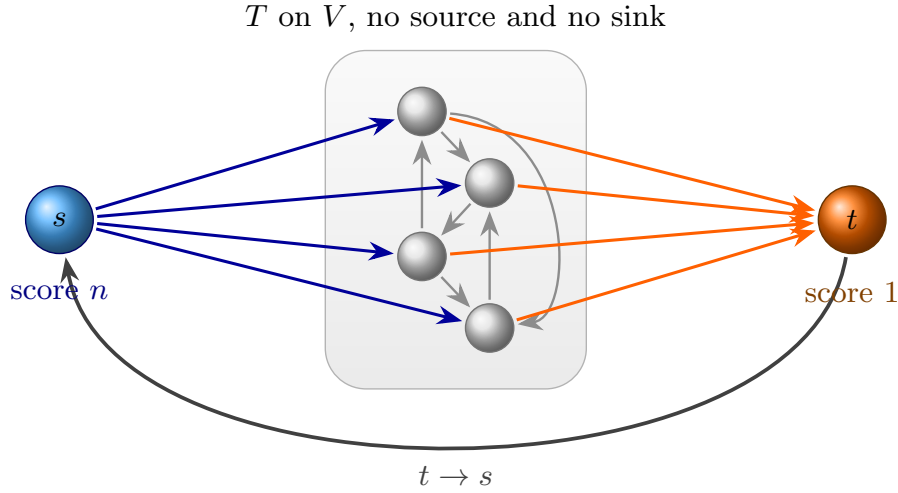


FIGURE 1. The tournament T^+ : s beats every vertex of T , every vertex of T beats t , and t beats s .

Lemma 2.1. *Let $n \geq 3$ and let T be a link-irregular tournament on n vertices with neither a source nor a sink. Then T^+ is a link-irregular tournament on $n + 2$ vertices with neither a source nor a sink.*

Proof. Since $1 \leq \text{sc}_T(v) \leq n - 2$ for $v \in V$, the scores in T^+ lie between 1 and n , while T^+ has $n + 2$ vertices; so T^+ has no source and no sink. We compare the links $T^+ - x$ for $x \in V \cup \{s, t\}$, each of which has $n + 1$ vertices, so that a source in it has score n .

The link of s . In $T^+ - s$ the vertex t has score 0, so $T^+ - s$ has a sink. Every $v \in V$ has score $\text{sc}_T(v) + 1 \leq n - 1$ there, so $T^+ - s$ has no source.

The link of t . In $T^+ - t$ the vertex s has score n , so $T^+ - t$ has a source. Every $v \in V$ has score $\text{sc}_T(v) \geq 1$ there, so $T^+ - t$ has no sink.

The link of $v \in V$. In $T^+ - v$ the vertex s has score $n - 1$, the vertex t has score 1, and $u \in V \setminus \{v\}$ has score $\text{sc}_{T-v}(u) + 1$, which lies between 1 and $n - 1$. So $T^+ - v$ has neither a source nor a sink.

Hence the link of s , the link of t and the links of vertices of V fall into three classes that no isomorphism can mix: the first has a sink but no source, the second a source but no sink, the third neither.

It remains to show $T^+ - v \not\cong T^+ - w$ for distinct $v, w \in V$. Call a vertex y of $T^+ - v$ *marked* if $\text{sc}(y) = 1$ and the unique out-neighbour of y has score $n - 1$. The vertex t is marked, its out-neighbour being s . Any other vertex of score 1 in $T^+ - v$ is some $u \in V \setminus \{v\}$ with $\text{sc}_{T-v}(u) = 0$; its only out-neighbour is t , whose score is $1 \neq n - 1$ because $n \geq 3$. So t is the only marked vertex of $T^+ - v$, and s is its out-neighbour. The same holds in $T^+ - w$. An isomorphism $\varphi: T^+ - v \rightarrow T^+ - w$ preserves scores and arcs, hence maps the marked vertex to the marked vertex, so $\varphi(t) = t$ and then $\varphi(s) = s$. Its restriction is an isomorphism from $T - v$ onto $T - w$, which is impossible because T is link-irregular. $\square$

3. TWO STARTING TOURNAMENTS

Let T_6 be the tournament on $\{1, \dots, 6\}$ with the arcs

$$1 \rightarrow 2, 3, 4, 5; \quad 2 \rightarrow 3, 4, 5, 6; \quad 3 \rightarrow 4, 6; \quad 4 \rightarrow 5, 6; \quad 5 \rightarrow 3; \quad 6 \rightarrow 1, 5,$$

where $i \rightarrow j, k$ means $i \rightarrow j$ and $i \rightarrow k$; the pairs not listed are oriented the other way. Its scores are 4, 4, 2, 2, 1, 2, so it has no source and no sink. Let T_7 be the tournament on $\{1, \dots, 7\}$ with the arcs

$$1 \rightarrow 2, 3, 4, 5, 6; \quad 2 \rightarrow 3, 4, 5, 6, 7; \quad 3 \rightarrow 4, 5, 7; \quad 4 \rightarrow 5, 6, 7; \quad 5 \rightarrow 6; \quad 6 \rightarrow 3; \quad 7 \rightarrow 1, 5, 6,$$

with scores 5, 5, 3, 3, 1, 1, 3; it has no source and no sink either.

Lemma 3.1. *T_6 and T_7 are link-irregular.*

Proof. Table 1 lists the score sequence of every vertex-deleted subtournament; each entry is obtained by deleting one vertex and recounting out-degrees. For T_7 the seven sequences are distinct, so the seven links are pairwise non-isomorphic. For T_6 the sequences are distinct except for $T_6 - 4$ and $T_6 - 5$, which both have score sequence (1, 1, 2, 3, 3). In $T_6 - 4$ the unique vertex of score 2 is 6, and it beats vertex 1, which has score 3. In $T_6 - 5$ the unique vertex of score 2 is 3, and it beats only the vertices 4 and 6, both of score 1. An isomorphism would carry the vertex of score 2 to the vertex of score 2 and preserve the scores of its out-neighbours, so $T_6 - 4 \not\cong T_6 - 5$. $\square$

TABLE 1. Score sequences (in increasing order) of the vertex-deleted subtournaments of T_6 and T_7 .

v	score sequence of $T_6 - v$	v	score sequence of $T_7 - v$
1	(1, 1, 2, 2, 4)	1	(1, 1, 2, 3, 3, 5)
2	(1, 2, 2, 2, 3)	2	(1, 1, 3, 3, 3, 4)
3	(0, 2, 2, 3, 3)	3	(0, 1, 3, 3, 4, 4)
4	(1, 1, 2, 3, 3)	4	(1, 1, 2, 3, 4, 4)
5	(1, 1, 2, 3, 3)	5	(1, 2, 2, 2, 4, 4)
6	(1, 1, 1, 3, 4)	6	(0, 2, 2, 3, 4, 4)
		7	(1, 1, 2, 2, 4, 5)

Proof of Theorem 1.1. Put $T_{n+2} = T_n^+$ for $n \geq 6$. Starting from T_6 and T_7 , Lemmas 3.1 and 2.1 show by induction on n that T_n is a link-irregular tournament on n vertices without a source or a sink, for every $n \geq 6$. For $2 \leq n \leq 5$ no link-irregular tournament exists, as Bastien and Khormali observed [2, Observation 7]: the n links are tournaments on $n - 1$ vertices, and up to isomorphism there are only 1, 1, 2, 4 tournaments on 1, 2, 3, 4 vertices, fewer than n in each case. $\square$

Remark 3.2. The construction is explicit: T_n is obtained from T_6 or T_7 by adding the pairs $(s_1, t_1), (s_2, t_2), \dots$ in turn, where s_i beats all earlier vertices except t_i , all earlier vertices beat t_i , and $t_i \rightarrow s_i$. The proof of the step uses the absence of a source and a sink in T , and this is exactly what it hands on to T^+ . Going through the 2^{15} tournaments on six vertices shows that T_6 is one of exactly four link-irregular tournaments on six vertices up to isomorphism, and that none of the four has a source or a sink. On seven vertices there are 139, eight of which have a source or a sink. Since an automorphism of a tournament maps each link onto an isomorphic link, a link-irregular tournament has no automorphism other than the identity.

Remark 3.3. Every link-irregular tournament on $n \geq 6$ vertices whose links have pairwise different score sequences gives such a tournament on $n + 2$ vertices with the same property, because the score sequence of $T^+ - v$ is that of $T - v$ with every entry raised by one and the entries 1 and $n - 1$ added. Starting from T_7 , this gives link-irregular tournaments of every odd order whose links are already told apart by their score sequences.

Use of generative AI. The author used Claude (Anthropic) for literature search, computation and drafting; the author checked the mathematics and takes full responsibility for the content.

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FORMERLY, ELECTRO-GRAVITATIONAL SPACE PROPULSION LABORATORY (EGSPL),
BHUBANESWAR, ODISHA 751030, INDIA

Email address: d.bhattacharjee@erl-forschung.de

Email address: itsdeep@live.com

URL: <https://orcid.org/0000-0003-0466-750X>